

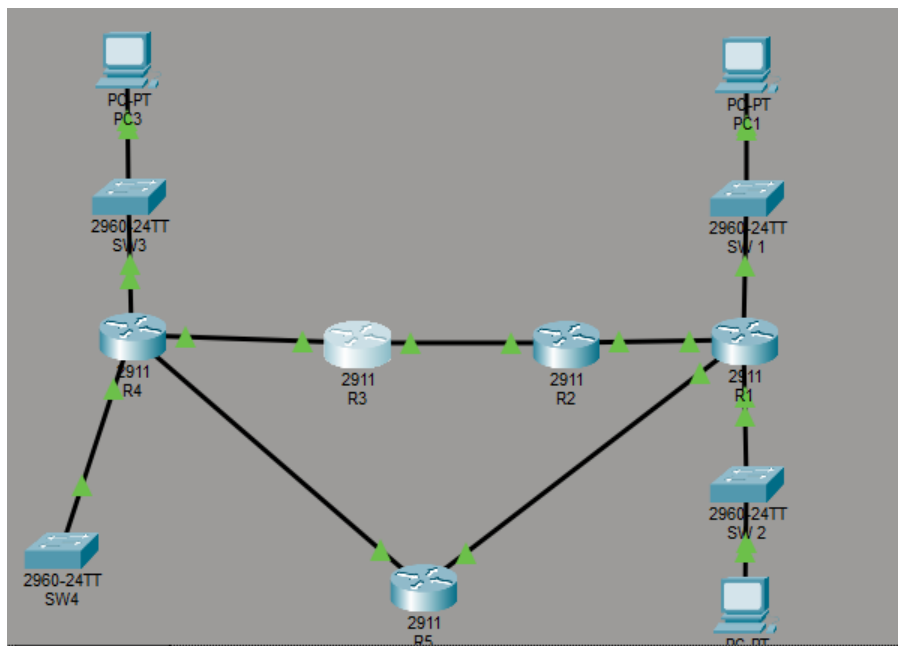


NET512 WEEK 8 PRAC

Lerato Modise



Screenshot of a topology



Configure RIPv1 on all Routers

```
R1>
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#
```

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router rip
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R2(config-router)#
```

```
R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router rip
R3(config-router)#network 10.0.0.0
R3(config-router)#no auto-summary
R3(config-router)#
```

```
R4>enable
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router rip
R4(config-router)#network 10.0.0.0
R4(config-router)#no auto-summary
R4(config-router)#
```

```

R5>enable
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router rip
R5(config-router)#network 10.0.0.0
R5(config-router)#no auto-summary
R5(config-router)#

```

Verify RIP v1 updates

```

R1#debug ip rip
RIP protocol debugging is on
R1#RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/0 (10.0.3.1)
RIP: build update entries
    network 10.1.0.0 metric 1
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/1 (10.1.0.1)
RIP: build update entries
    network 10.0.3.0 metric 1
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/0 (10.0.3.1)
RIP: build update entries
    network 10.1.0.0 metric 1
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/1 (10.1.0.1)
RIP: build update entries
    network 10.0.3.0 metric 1
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/0 (10.0.3.1)
RIP: build update entries
    network 10.1.0.0 metric 1
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/1 (10.1.0.1)
RIP: build update entries
    network 10.0.3.0 metric 1

R1#undebug all
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/0 (10.0.3.1)
RIP: build update entries
    network 10.1.0.0 metric 1
RIP: sending v1 update to 255.255.255.255 via GigabitEthernet0/1 (10.1.0.1)
RIP: build update entries
    network 10.0.3.0 metric 1

All possible debugging has been turned off

```

Upgrade to RIP v2

```

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#

```

```

R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#

```

```

R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#

```

```

R4>enable
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#

```

```

R5>enable
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#

```

Confirm RIP v2 updates

```
R1#debug ip rip
RIP protocol debugging is on
R1#RIP: sending v2 update to 224.0.0.9 via GigabitEthernet0/0 (10.0.3.1)
RIP: build update entries
      10.1.0.0/24 via 0.0.0.0, metric 1, tag 0
RIP: sending v2 update to 224.0.0.9 via GigabitEthernet0/1 (10.1.0.1)
RIP: build update entries
      10.0.3.0/24 via 0.0.0.0, metric 1, tag 0

R1#RIP: sending v2 update to 224.0.0.9 via GigabitEthernet0/0 (10.0.3.1)
RIP: build update entries
      10.1.0.0/24 via 0.0.0.0, metric 1, tag 0
RIP: sending v2 update to 224.0.0.9 via GigabitEthernet0/1 (10.1.0.1)
RIP: build update entries
      10.0.3.0/24 via 0.0.0.0, metric 1, tag 0

R1#undebug all
All possible debugging has been turned off
R1#
```

Verify RIP routes

```
R1#show ip route rip
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
```

View RIP database

```
R1#show ip rip database
10.0.3.0/24      auto-summary
10.0.3.0/24      directly connected, GigabitEthernet0/0
10.1.0.0/24      auto-summary
10.1.0.0/24      directly connected, GigabitEthernet0/1
R1#
```

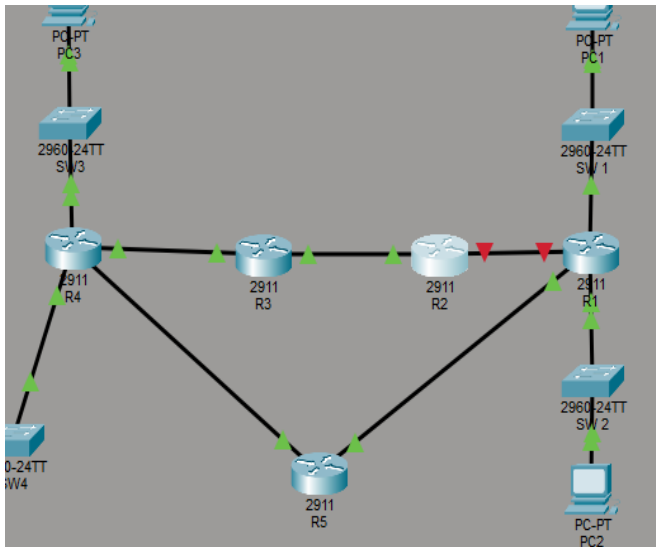
```
R2#show ip rip database
10.0.0.0/24      auto-summary
10.0.0.0/24      directly connected, GigabitEthernet0/0
R2#
```

```
R3#show ip rip database
10.0.0.0/8       auto-summary
10.0.0.0/8       directly connected, GigabitEthernet0/0
R3#
```

```
R4>enable
R4#show ip rip database
10.1.1.0/24      auto-summary
10.1.1.0/24      directly connected, GigabitEthernet0/2
10.1.3.0/24      auto-summary
10.1.3.0/24      directly connected, GigabitEthernet0/1
R4#
```

```
R5#show ip rip database
10.0.3.0/24      auto-summary
10.0.3.0/24      directly connected, GigabitEthernet0/1
10.1.3.0/24      auto-summary
10.1.3.0/24      directly connected, GigabitEthernet0/0
R5#
```

Test ospf Behaviour



Router 1 also turned down after turning down router 2

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
```

Compare databases

```
R1#show ip ospf database
R1#show ip rip database
10.0.3.0/24    auto-summary
10.0.3.0/24    directly connected, GigabitEthernet0/0
10.1.0.0/24    auto-summary
10.1.0.0/24    directly connected, GigabitEthernet0/1
R1#
R1#
```

Reflection Questions

RIP v1 vs RIP v2 updates:

RIP v1 uses broadcast; RIP v2 uses multicast (224.0.0.9)

RIP v2 supports subnet masks and authentication